

ARTIFICIAL INTELLIGENCE

CO3 – ASSESSMENT TOOL 3

Resolution Algorithm Implementation

Course Name	Artificial Intelligence
Course Code	CSA17
Questions	5
Total Marks	30

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Resolution Algorithm – Assignment Answers

The following five problems are solved using propositional logic, CNF conversion, goal negation, resolution steps, and the empty clause ($\square$), following the questions in the uploaded assessment.

Question 1 – Rain and Wet Ground

Knowledge Base

- If it rains, then the ground becomes wet.
- It is raining.

Goal

Prove that the ground is wet using the Resolution Algorithm.

1. Propositional Logic

Let R = It is raining. Let W = The ground is wet.

Rule: $R \rightarrow W$

Fact: R Goal: W

2. CNF Conversion

$R \rightarrow W \equiv \neg R \vee W$

CNF Clauses:

- $C1: \neg R \vee W$
- $C2: R$

3. Negate the Goal

Negated goal: $\neg W$

4. Resolution Steps

Resolve $C1 (\neg R \vee W)$ with $C2 (R)$:

Result: W

Now resolve W with the negated goal $\neg W$:

$$\begin{array}{r} W \quad \neg W \\ \backslash \quad / \\ \square \end{array}$$

Empty clause $\square$ is obtained.

Final Conclusion

The goal W (the ground is wet) is proved by Resolution.

Question 2 – Student Assignment Submission

Knowledge Base

- If a student submits the assignment, then the student receives internal marks.
- Rahul submitted the assignment.

Goal

Prove that Rahul receives internal marks using the Resolution Algorithm.

1. Propositional Logic

Let S = Rahul submitted the assignment. Let M = Rahul receives internal marks.

Rule: $S \rightarrow M$

Fact: S

Goal: M

2. CNF Clauses

- C1: $\neg S \vee M$
- C2: S

3. Negate the Goal

Negated goal: $\neg M$

4. Resolution Steps

Resolve C1 ($\neg S \vee M$) with C2 (S):

Result: M

$$\begin{array}{cc} M & \neg M \\ \backslash & / \\ \square \end{array}$$

The derived M and the negated goal $\neg M$ produce the empty clause $\square$.

Final Conclusion

Rahul receives internal marks is proved by Resolution.

Question 3 – Library Membership

Knowledge Base

- If a person is a library member, then the person can borrow books.
- Priya is a library member.

Goal

Prove that Priya can borrow books using the Resolution Algorithm.

1. Propositional Logic

Let L = Priya is a library member. Let B = Priya can borrow books.

Rule: $L \rightarrow B$

Fact: L

Goal: B

2. CNF Clauses

- C1: $\neg L \vee B$
- C2: L

3. Negate the Goal

Negated goal: $\neg B$

4. Resolution

Resolve C1 ($\neg L \vee B$) with C2 (L):

Result: B

Now resolve B with $\neg B$:

$$\begin{array}{cc} B & \neg B \\ \backslash & / \\ & \square \end{array}$$

No contradiction remains after removing the negated goal; the empty clause proves the goal.

Final Conclusion

Priya can borrow books is proved using Resolution.

Question 4 – Placement Eligibility

Knowledge Base

- If a student clears the aptitude test, then the student is eligible for placement.
- Arun cleared the aptitude test.

Goal

Prove that Arun is eligible for placement using the Resolution Algorithm.

1. Propositional Logic

Let A = Arun cleared the aptitude test. Let E = Arun is eligible for placement.

Rule: $A \rightarrow E$

Fact: A Goal: E

2. CNF Clauses

- C1: $\neg A \vee E$
- C2: A

3. Negate the Goal

Negated goal: $\neg E$

4. Resolution Steps

Resolve C1 ($\neg A \vee E$) with C2 (A):

Result: E

Resolve E with $\neg E$:

$$\begin{array}{cc} E & \neg E \\ \backslash & / \\ & \square \end{array}$$

5. Resolution Tree

$$\begin{array}{cc} \neg A \vee E & A \\ \backslash & / \\ E & \\ | & \\ \neg E & \\ | & \\ \square & \end{array}$$

Final Conclusion

Arun is eligible for placement. The empty clause $\square$ confirms that the goal is proved.

Question 5 – Access Control System

Knowledge Base

- If a user enters the correct password, then the user is authenticated.
- If a user is authenticated, then the user is granted access.
- The user entered the correct password.

Goal

Prove that the user is granted access using the Resolution Algorithm.

1. Propositional Logic

Let P = User entered the correct password.

Let A = User is authenticated.

Let G = User is granted access.

Rules: $P \rightarrow A$; $A \rightarrow G$

Fact: P

Goal: G

2. CNF Conversion

- $C1: \neg P \vee A$
- $C2: \neg A \vee G$
- $C3: P$

3. Negate the Goal

Negated goal: $\neg G$

4. Resolution Steps

Step 1: Resolve $C1 (\neg P \vee A)$ with $C3 (P)$.

Result: A

Step 2: Resolve $C2 (\neg A \vee G)$ with A .

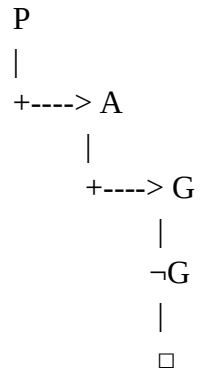
Result: G

Step 3: Resolve G with the negated goal $\neg G$.

$$\begin{array}{r} G \quad \neg G \\ \backslash \quad / \\ \square \end{array}$$

The empty clause $\square$ is obtained.

5. Resolution Chain



Final Conclusion

The user is granted access. The Resolution Algorithm derives the empty clause $\square$, so the goal is proved.

FINAL ANSWERS SUMMARY

Question	Conclusion
Q1 – Rain and Wet Ground	Ground is wet – PROVED
Q2 – Student Assignment	Rahul receives internal marks – PROVED
Q3 – Library Membership	Priya can borrow books – PROVED
Q4 – Placement Eligibility	Arun is eligible for placement – PROVED
Q5 – Access Control	User is granted access – PROVED